## *Chapter 8: Introduction to Environmental Sanitation*

**Definition of Sanitation:**

Sanitation means keeping our surroundings clean and healthy. It helps prevent diseases and protect the environment. It includes the safe management of water and waste. The main activities of sanitation are:

1. Providing safe and clean water.
2. Removing dirty water properly.
3. Disposing of human waste safely.
4. Removing garbage quickly.

In simple words: Sanitation helps us live in a clean and healthy environment and reduces the risk of diseases.

**Two Main Objectives of Sanitation:**

1. To improve public health: Proper sanitation helps keep people healthy. It prevents the spread of diseases by safely removing human waste and reducing contact with harmful germs.
2. To reduce environmental pollution Proper sanitation helps keep water, soil, and our surroundings clean. It prevents waste from polluting the environment and protects nature.

**Name five routes through which sanitation-related infectious diseases can spread.**

Sanitation-related infectious diseases can spread through the following five routes:

1. Contaminated food or water: Eating contaminated food or drinking dirty water.
2. Infected beef: Eating beef containing tapeworms.
3. Contaminated water: Coming into contact with polluted water.
4. Contaminated soil: Touching soil containing harmful germs or worms.
5. Insects: Flies and other insects carry germs from faeces to food.

**Classify the different types of waste.**

Waste can be classified into different types according to its source and nature. The main types are:

1. Human waste or excreta: It includes human faeces and urine.
2. Municipal sewage: It is wastewater carried through sewers. It may include household waste, industrial waste, and rainwater.
3. Domestic sewage: It is wastewater from toilets, bathrooms, kitchens, and sinks in homes and buildings.
4. Sullage: It is wastewater from kitchens and washbasins that does not contain human faeces or urine.
5. Industrial waste: It is waste produced by factories during manufacturing and other industrial activities.
6. Stormwater: It is rainwater that flows over roads and land.
7. Solid waste: It includes unwanted solid materials such as food waste, paper, plastic, glass, and metal.

## Sanitation Systems

A sanitation system collects, stores, treats, and safely disposes of human waste to protect health and the environment. Main functions:

1. Excretion and storage: Releasing and storing waste.
2. Collection and transportation: Collecting and moving waste.
3. Processing and treatment: Treating waste to make it safer.
4. Disposal and recycling: Safely disposing of or reusing waste.

## Classify the different types of sanitation systems.

Sanitation systems can be classified according to how waste is collected, transported, treated, and disposed of. The main types are:

1. On-site sanitation system: In this system, waste is collected, treated, and disposed of at the place where it is produced. Example: Pit latrine.

2. Off-site sanitation system: In this system, waste is collected and transported to another place for treatment and disposal. Example: Conventional sewerage system.

3. Dry sanitation system; In this system, water is not used to flush or dilute the waste. It is commonly used in places without a piped water supply. Example: Pit latrine.

4. Wet sanitation system; In this system, water is used to flush or dilute the waste. It is suitable for places where sufficient water is available. Example: Flush toilet connected to a sewerage system.

## Differentiate between on-site and off-site sanitation systems.

On-site and off-site sanitation systems are different in the way they collect and manage human waste.

| On-site sanitation system | Off-site sanitation system |
|---|---|
| 1. Waste is managed where it is produced. | 1. Waste is carried to another place for treatment and disposal. |
| 2. A separate sewer network is generally not required. | 2. A sewer network or another waste transportation method is required. |
| 3. It is often suitable for rural areas with low population density. | 3. It is often suitable for densely populated urban areas. |
| 4. It may require less transportation infrastructure. | 4. It needs suitable facilities for collecting and transporting waste. |
| 5. Example: Pit latrine. | 5. Example: Conventional sewerage system. |

## Differentiate between dry and wet sanitation systems.

Dry and wet sanitation systems differ mainly in their use of water.

| Dry sanitation system | Wet sanitation system |
|---|---|
| 1. Water is not used to flush or dilute waste. | 1. Water is used to flush or dilute waste. |
| 2. It is useful in areas without a piped water supply. | 2. It is suitable for areas with a sufficient water supply. |

| | |
|---|---|
| 3. It is often used where sewerage facilities are not available. | 3. It may be used with septic tanks or conventional sewerage systems. |
| 4. Example: Pit latrine. | 4. Example: Flush toilet system. |

**Explain the appropriateness of sanitation systems.**

The appropriateness of a sanitation system means selecting a system that is suitable for the people and the area. The choice depends on the following factors:

1. People's needs: The system should protect health and provide privacy and convenience.

2. Water supply: Dry systems may be suitable where water is limited, while flush toilets require sufficient water.

3. Population density: On-site systems are often suitable for rural areas, while off-site systems may be more suitable for crowded urban areas.

4. Maintenance: Toilets must be properly used, regularly cleaned, and connected to a suitable waste disposal system.

5. Environmental protection: The system should prevent pollution of groundwater, rivers, and the surrounding environment.

Therefore, the best sanitation system is one that meets people's needs, suits local conditions, and protects public health and the environment.

**What are the important factors for sanitation in Bangladesh?**

The important factors for sanitation in Bangladesh are:

1. Housing Density: The number of people living in an area affects the type of toilet needed. Pit latrines are suitable for many rural areas.

2. Water Supply: If enough water is available, flush toilets can be used. If water is limited, dry toilets may be better.

3. Problems with Pit Latrines: Floods, soft soil, rocky ground, and high groundwater can make pit latrines difficult to build.

4. Operation and Maintenance: Toilets should be cleaned regularly and repaired when needed.

5. Soil Permeability: It means how easily water passes through the soil. If water cannot pass through easily, the pit may overflow.

6. Groundwater Pollution: Toilet waste can pollute underground water and nearby wells. Toilets should be built in suitable places to prevent pollution.

**simple pit latrine**

A simple pit latrine is a low-cost sanitation system consisting of a pit dug into the ground, a squatting slab, and a superstructure. Human waste enters the pit through a squatting hole. The liquid portion infiltrates into the surrounding soil, while solid materials remain in the pit and gradually decompose.

| Advantages: | Disadvantages: |
|---|---|
| - Low construction cost. | Flies may breed in the pit. |
| - Easy to construct and maintain. | Bad odour may develop. |
| - Reduces hookworm transmission. | Flies may spread diseases. |
| - Provides improved sanitation. | Poor construction may cause collapse. |

VIP latrine

A Ventilated Improved Pit (VIP) latrine is an improved form of a simple pit latrine designed to control bad odour and flies. It consists of a pit, a cover slab, a superstructure, a vent pipe and a fly screen.

Working Principle: Fresh air enters through the squatting hole and carries bad odour through the vent pipe. Flies move toward the light at the top of the vent pipe but are trapped by the fly screen.

Advantages:

- Controls odour and insects.
- Low construction cost.
- Easy maintenance.
- Minimum water requirement.
- Provides improved sanitation.

Disadvantages:

- Risk of groundwater pollution.
- Requires space for pit relocation.
- Difficult construction in rocky or high groundwater areas.

Compost Latrine?

A compost latrine is a sanitation system where human excreta (faeces) and organic materials, such as vegetable waste, are decomposed by microorganisms to produce compost that can be used as a soil conditioner or fertilizer.

The basic principle is aerobic decomposition, which means decomposition in the presence of oxygen.

For proper composting, the following conditions are important:

- Suitable Carbon-to-Nitrogen (C/N) ratio
- Low moisture content
- Availability of air (oxygen)
- Proper temperature and sufficient decomposition time

Continuous Compost Latrine

Continuous compost latrines, also called Multrum-type latrines, allow waste to be added continuously while decomposed material is removed periodically.

Working process:

1. Human excreta and vegetable waste are deposited into a receptacle.

2. The mixture moves down a sloping floor toward a storage chamber.

3. Air enters the chamber and helps microorganisms decompose the waste.

4. The material remains inside for several years to become compost.

5. The compost is removed through the storage chamber.

For effective composting, a C/N ratio of about 30:1 is suggested.

Batch Compost Latrine:

A batch compost latrine is commonly known as a Double Vault Compost Latrine.

It has two watertight chambers (vaults), which are used alternately.

Working process:

1. Human excreta and dry organic materials are deposited into the first chamber.
2. The first chamber is used until it is nearly full.
3. It is then sealed, and the second chamber is used.
4. The waste in the first chamber is allowed to decompose for at least one year.
5. The compost is removed and may be used as fertilizer when properly treated.

Materials such as ash, vegetable matter, soybean leaves and sawdust are used to help maintain the required C/N ratio.

**Advantages of Compost Latrines**

1. Suitable for areas where human excreta are traditionally used on land.
2. Compost can be used as fertilizer when properly processed.

3. Requires little water or no flushing water.
4. Can be constructed in rocky areas, where pit excavation is difficult.
5. Can reduce the risk of contaminating vulnerable water supplies.

Disadvantages of Compost Latrines

1. Requires additional biodegradable organic matter to maintain the correct C/N ratio.
2. Requires careful and regular operation.
3. Not suitable where people prefer water for anal cleansing.
4. May attract flies and cause unpleasant odours if not properly maintained.
5. Pathogenic organisms may remain if waste is not stored or treated long enough.

Introduction to Pour-Flush Sanitation

Pour-flush sanitation is an improved form of the simple pit latrine. It uses a small amount of water to flush human excreta into a pit.

The main difference between a simple pit latrine and a pour-flush latrine is the water seal, which prevents unpleasant smells and insects from entering the toilet.

According to your PDF, a pour-flush latrine has three main components:

1. Superstructure – provides privacy and protection.
2. Pan with water seal – allows flushing and prevents odours.
3. Pit – stores and decomposes human waste.

Direct Pit Pour-Flush Latrine

This is a modified simple pit latrine. The squatting pan is placed directly above the pit, and a water seal is installed. The water seal is approximately 25 mm deep.

Working process:

1. Human excreta are deposited in the pan.
2. Water is poured into the pan for flushing.
3. Waste flows directly into the pit.
4. Wastewater infiltrates into the surrounding soil.
5. Solid waste accumulates and decomposes inside the pit.

Off-set Pit Pour-Flush Latrine

An off-set pit pour-flush latrine has a pit located away from the toilet pan. The pan and pit are connected by a pipe of approximately 100 mm diameter.

Working process:

1. Human excreta are flushed using water.

2. Waste passes through the water seal.
3. It flows through the connecting pipe into the pit.
4. Solid waste accumulates, while the liquid infiltrates into the soil.

Alternating Twin Off-set Pit Pour-Flush Latrine

This system consists of one squatting pan, two separate pits and a Y-junction that directs wastewater into either pit.

Working process:

1. Waste initially enters the first pit.
2. When the first pit becomes full, its connection is closed.
3. Waste is then directed into the second pit.
4. Waste in the first pit decomposes during the resting period.
5. After approximately 18–24 months, the decomposed contents may be removed and the pit reused.

This system is particularly useful for long-term sanitation.

Functions of Pour-Flush Latrines

According to the PDF, pour-flush latrines perform two important functions.

Function 1: Flushing and Water Seal

- About 2–3 litres of water are used for flushing.
- Some water remains inside the trap.
- This water forms a barrier against odours and insects.
- It provides a more hygienic toilet environment.

Function 2: Waste Treatment and Disposal

- Approximately 5–10 litres per person per day of wastewater (including excreta, flushing and anal-cleansing water) may enter the pit.
- The liquid portion infiltrates into the surrounding soil.
- Solids accumulate and undergo digestion.
- The system reduces the need for conventional sewerage.

Design Considerations for Pour-Flush Latrines

1. Pit shape: Circular, square, rectangular or triangular.
2. Pit depth: In most cases in Bangladesh, pits should not exceed 1.8 metres.
3. Pit lining: Can be made of burnt clay, concrete, brick masonry or bamboo.
4. Minimum space: Keep about 0.5 metres of free space above the waste in the pit.
5. Groundwater safety: The pit bottom should be at least 0.5 metres above the highest groundwater level according to this section.

6. Distance from water sources: Maintain at least 10 metres from tubewells or other water bodies.
7. Connecting pipe: Approximately 100 mm diameter, with a recommended slope between 1:50 and 1:20

Advantages of Pour-Flush Sanitation

- Less expensive than conventional sewerage.
- More appropriate and hygienic excreta disposal.
- Requires a small quantity of flushing water (about 2–3 litres).
- Can be upgraded to a sewerage system or septic tank.
- Eliminates unpleasant odours and flies.
- Safe for children.
- Can be located inside the house.
- Potential for using treated pit contents as a soil conditioner.
- Single-pit systems are relatively easy to construct and maintain.

Disadvantages of Pour-Flush Sanitation

- Requires separate sludge disposal facilities.
- Needs at least 5 litres of water per person per day throughout the year.
- Water seal may become clogged if garbage is thrown into it.
- Construction is difficult and expensive in areas with high groundwater levels and shallow soil over rock.
- May pollute nearby water sources.

Prevention of Groundwater Pollution

Liquid from pit latrines can carry microorganisms into groundwater. The textbook cites research indicating that a vertical separation of at least 2 metres between the pit bottom and the groundwater table, together with a horizontal distance of 10 metres from a drinking-water well, can significantly reduce microbial contamination risks under certain soil conditions.

Aqua Privies

An Aqua Privy is an on-site sanitation system consisting of a squatting pan or seat, a vertical drop pipe, and a watertight tank containing water. Human excreta passes through the drop pipe into the water-filled tank, where it is decomposed by anaerobic bacteria (bacteria that work without oxygen). The main difference between an Aqua Privy and a simple pit latrine is that an Aqua Privy uses a water-filled tank instead of a dry pit.

Working Principle of Aqua Privy

 Human excreta enters the toilet pan.

 Excreta drops through the vertical pipe into the water-filled tank.

 The bottom of the drop pipe remains below the water level, forming a water seal that helps prevent odours and insects from entering the toilet.

 Solids settle and are decomposed anaerobically.

 The liquid effluent flows out through the outlet and requires further disposal or treatment.

 Accumulated sludge must be removed periodically.

Water Seal and Maintenance

Maintaining the water level is the most important requirement of an Aqua Privy.

- The bottom of the drop pipe should remain at least 100 mm below the water level.
- The system requires regular maintenance and cleaning.
- If the water seal is lost, odours and insects may enter the toilet.
- A modified design may include two interconnected compartments, one for sludge collection and the other for liquid discharge.

Advantages of Aqua Privies

 Low odour and insect problems if the water seal is maintained.

 Minimal health risks.

 Low annual costs.

 Potential for upgrading.

 Potential for sludge disposal.

Disadvantages of Aqua Privies

- Difficulties maintaining the water level and water seal.
- Higher investment costs than pour-flush or VIP latrines.
- Cannot be installed within the house.
- Tanks need to be watertight.
- Additional water is needed to maintain the water level.
- Regular desludging is required, usually by local municipal authorities.

Septic Tank

A septic tank is a buried, watertight tank used to collect and partially treat domestic wastewater. It separates solid waste from liquid waste and allows organic matter to decompose through anaerobic digestion (decomposition without oxygen). The treated liquid, called effluent, leaves the septic tank and requires further treatment or safe disposal.

Processes in a Septic Tank

Three major processes take place inside a septic tank:

1. Separation and Sedimentation: Heavy solids settle at the bottom, while light materials float to the surface.
2. Anaerobic Digestion: Anaerobic bacteria break down organic matter in the sludge and scum.
3. Sludge and Scum Storage: The remaining solids are stored until they are removed through desludging.